

Endemic Amazonian core and episodic coastal expansion of dengue in Peru, 2018–2024: a spatio-temporal persistence analysis at the province level

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Abstract

Peru recorded the largest dengue epidemics in its recent history during 2023 and 2024. Most spatial studies of dengue apply local cluster statistics to a single year or to cumulative incidence, which prevents permanent transmission foci from being distinguished from foci that activate episodically during epidemic cycles. We analysed the spatial distribution, temporal evolution and persistence of dengue incidence clusters across the 196 provinces of Peru during 2018–2024. Notified cases, official population projections and provincial cartography were integrated into a balanced panel of 196 provinces over seven years (1372 observations). The Global Moran’s Index, the Getis–Ord G_i^* statistic and Local Indicators of Spatial Association were applied to every year of the period. Given the marked skewness of provincial incidence (coefficients between 2.43 and 7.33), statistics were computed on log-transformed incidence with conditional permutation inference (9,999 permutations) and false discovery rate correction; robustness was assessed under four alternative specifications. National incidence rose from 14.88 cases per 100,000 inhabitants in 2018 to 797.72 in 2024, while provinces reporting no cases fell from 134 to 91. Global Moran’s I was positive and significant in every year and remained stable between 0.525 and 0.644; the oscillation observed on the original scale proved to be an artefact of distributional skewness rather than evidence of spatial reorganisation. Nine provinces in Loreto and San Martín were significant hotspots

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in all seven consecutive years, and eleven provinces of those departments in all five years of the 2020–2024 subperiod. The coastal component reached its maximum extent in 2023 and showed no historical persistence, with Sullana as the sole exception. In 2024, 23 hotspots and 31 coldspots were identified, the latter concentrated entirely in the southern and central Andean highlands. Dengue distribution in Peru follows a clustered spatial structure that is stable over time and organised into two components with distinct dynamics: a permanent endemic Amazonian core and an episodic coastal expansion. This distinction, not identifiable through cross-sectional analyses, supports geographically differentiated surveillance and vector control strategies.

Keywords: dengue; spatial analysis; spatial autocorrelation; disease clusters; hotspot detection; epidemiological surveillance; Peru

1 Introduction

Dengue is one of the most important vector-borne viral diseases worldwide. It is caused by any of the four serotypes of the dengue virus (DENV-1 to DENV-4) and is transmitted mainly by *Aedes aegypti*, although *Aedes albopictus* may also contribute to transmission in some regions. Over recent decades its incidence has risen steadily as a consequence of the geographical expansion of its vectors, unplanned urban growth, increased human mobility and the effects of climate change, all of which have favoured outbreaks in previously unaffected areas. Approximately half of the world population is currently at risk of infection, and the disease is endemic in more than one hundred countries (Bhatt et al., 2013; World Health Organization, 2024).

In Latin America, dengue represents a major public health problem owing to climatic conditions favourable to vector proliferation, rapid urbanisation processes and limitations in epidemiological control strategies. In recent years several countries in the region have experienced large-magnitude epidemics, with historical increases in case numbers and expansion of the disease into areas where sustained transmission had not previously occurred (Organización Panamericana de la Salud, 2024; Ryan et al., 2019).

In Peru, dengue has increased steadily in recent years and has become one of the vector-borne diseases with the greatest impact on the national health system. Climatic variability, the El Niño phenomenon, disorderly urban growth, deficient water supply and domestic water storage favour the proliferation of *Aedes aegypti* and contribute to the persistence of epidemic outbreaks in various regions of the country. The years 2023 and 2024 recorded the highest transmission levels documented in the country's recent history, showing that the disease maintains a wide territorial distribution and represents a permanent challenge for health authorities (Centro Nacional de Epidemiología, Prevención y Control de Enfermedades, 2024; Ministerio de Salud del Perú, 2024).

Dengue exhibits marked spatial heterogeneity, since transmission does not occur at random but is influenced by environmental, climatic, demographic and socioeconomic factors that vary between neighbouring territories. In this context, spatial statistical techniques allow the identification of spatial dependence patterns, high- and low-risk clusters, and priority areas for public health intervention. Tools such as the Global Moran's Index, Local Indicators of Spatial Association (LISA) and the Getis-Ord G_i^* statistic have been widely used to detect spatial autocorrelation and locate transmission foci (Anselin, 1995; Bivand et al., 2013; Ord and Getis, 1995). A recent scoping review concludes that these three statistics are currently the most widely

employed tools for studying the spatial distribution of dengue worldwide (Prakash Nayak et al., 2025).

The evidence available for Peru has advanced substantially in recent years. Rufasto Goche et al. (2025) analysed the epidemiological dynamics of dengue between 2000 and 2022 at the departmental scale and concluded that Loreto, Ucayali, Piura and San Martín persistently concentrated the highest disease burden. At the local scale, studies conducted in Iquitos by Morrison et al. (2010) and Liebman et al. (2012) documented marked spatial heterogeneity of transmission and the existence of stable foci associated with vector distribution and population mobility. These antecedents firmly establish that the distribution of dengue in the country is not random.

Three limitations nevertheless persist in the available evidence. First, most studies applying local indicators of spatial association do so for a single year or for the cumulative incidence of the period, which prevents permanent transmission foci from being distinguished from those that activate episodically during epidemic cycles. This distinction is relevant for health planning, since the two types of focus call for different intervention strategies. Second, dengue incidence displays a strongly skewed distribution at the provincial scale, a condition whose effect on the behaviour of autocorrelation statistics is rarely evaluated, even though it may condition the interpretation of their temporal variations. Third, the application of multiple-comparison corrections, necessary when hundreds of spatial units are assessed simultaneously, is not a widespread practice in this literature.

Against this background, the aim of the present study was to analyse the spatial distribution, temporal evolution and persistence of dengue incidence clusters across the 196 provinces of Peru during the period 2018–2024. Specifically, we assessed the existence of global spatial autocorrelation using Moran’s Index and its robustness to the measurement scale, the neighbourhood criterion and the small-numbers problem; identified high- and low-incidence clusters using the Getis–Ord G_i^* statistic and Local Indicators of Spatial Association, applied to every year of the period and with correction for multiple comparisons; and characterised the temporal persistence of the identified foci in order to distinguish endemic from epidemic components of transmission. The results aim to contribute evidence supporting the strengthening of epidemiological surveillance and the design of geographically differentiated prevention and control strategies.

2 Materials and Methods

2.1 Study area

The study area comprises the 196 provinces of Peru, distributed across 24 departments and the Constitutional Province of Callao. Owing to its wide geographical, climatic and ecological diversity, the Peruvian territory presents heterogeneous conditions that influence the distribution of dengue and the presence of *Aedes aegypti*. Provinces constitute the spatial unit of analysis, allowing the territorial variability of incidence to be assessed and clustering patterns to be detected at the national level. The analysis covered the years 2018 to 2024, in order to evaluate the spatial evolution of dengue before, during and after the 2023 epidemic.

2.2 Study design

This is a quantitative, applied study with an observational, ecological and longitudinal design. An ecological approach was adopted because the unit of analysis was the province rather than the individual, while the longitudinal character allowed the spatial evolution of dengue incidence to be assessed over the 2018–2024 period.

The analysis was based on secondary information from official Peruvian government sources. Dengue cases were aggregated at the provincial level and integrated with demographic and cartographic information to build a spatial database, on which epidemiological indicators were computed and spatial statistical techniques applied.

2.3 Workflow and reproducibility

The research followed a sequential, modular workflow organised into numbered directories reproducing the logical order of the analysis: (i) retrieval and storage of the original unmodified data; (ii) cleaning, standardisation and integration of the databases; (iii) incorporation of the provincial vector cartography; (iv) development of the analysis scripts; (v) generation of numerical results; (vi) production of thematic maps; (vii) export of tables; (viii) export of figures; and (ix) manuscript preparation.

This organisation ensures that the original data remain unaltered, that each processing stage is documented in an executable script, and that the whole analysis can be reproduced from the primary sources. All permutation-based procedures were run with a fixed random seed.

2.4 Data sources

The data used came from three official Peruvian government sources (Table 1). First, we used the dengue epidemiological surveillance database for the period 2000–2024, published by the National Center for Epidemiology, Disease Prevention and Control (CDC Peru) of the Ministry of Health through its open data platform (Ministerio de Salud del Perú, 2024). Records between 2018 and 2024 were extracted, considering the provincial-level geographical location and the year of notification of each case.

The surveillance database was downloaded on 20 July 2026 from the National Open Data Portal of Peru. The download date is reported because the published figures are subject to periodic reclassification: for 2023, for example, the official figures released range between 256,641 and 273,172 cases depending on the cut-off date considered.

Second, we used the database “Peru: Total Population Projected to 30 June of each year, by department, province and district, 2018–2026”, published by the National Institute of Statistics and Informatics (Instituto Nacional de Estadística e Informática, 2024b), which provided the projected population of each province for computing incidence per 100,000 inhabitants.

Finally, we used the official vector cartography of the provinces of Peru (Instituto Nacional de Estadística e Informática, 2024a), containing the provincial administrative boundaries used to build the spatial database and produce the thematic maps.

Table 1: Data sources used in the study.

Source	Information used	Period
MINSA – CDC Peru	Notified dengue cases	2018–2024
INEI	Projected population by province	2018–2024
INEI – National Geostatistical Framework	Provincial boundaries (shapefile)	Current

2.5 Data preparation and integration

The three sources underwent cleaning, standardisation and integration. The dengue case database was first filtered to retain only records for 2018–2024. Cases were then aggregated at the provincial level using the ubigeo code and the year of notification as grouping variables.

The surveillance database is structured with one record per notified case and includes both laboratory-confirmed and probable cases, in their three clinical forms: dengue without warning signs (ICD-10 A97.0), dengue with warning signs (A97.1) and severe dengue (A97.2). The 2018–2024 period comprised 704,097 records, distributed across the 52 or 53 epidemiological weeks of

each year, with no truncation in the final year.

Records do not include a personal identifier, so two cases sharing department, province, district, locality, clinical form, year, epidemiological week, age and sex are indistinguishable from one another. Consequently, no duplicate-removal procedure was applied: identical records correspond to distinct individuals with coinciding characteristics, and their suppression would lead to substantial underestimation of case counts. The total obtained under this criterion is consistent with the official figures released by CDC Peru for the period.

The geographical location of each case was established from the district ubigeo field of the surveillance database, which is distinct from the field identifying the notifying regional health directorate. The district ubigeo was converted to a provincial ubigeo by retaining the first four digits of the code. All provincial ubigeos present in the case database matched a polygon in the official cartography, with no orphan records and no loss of cases during integration. During the study period, 124 of the 196 provinces notified at least one case.

In parallel, the population projection database was processed to select provincial records for the same period. Ubigeo codes and province names were standardised to ensure compatibility across databases. To maintain a homogeneous structure, a balanced panel was built comprising the 196 provinces and the seven study years, yielding 1372 spatial observations. Records with no notified cases were treated as zero values.

2.6 Computation of dengue incidence

To allow comparisons between provinces of different population sizes, annual dengue incidence was computed using the projected population for each province and year. For each province i and year t :

$$\text{Incidence}_{it} = \frac{\text{Cases}_{it}}{\text{Population}_{it}} \times 100,000, \quad (1)$$

where Cases_{it} is the annual number of notified dengue cases in province i during year t and Population_{it} is the corresponding projected population.

2.7 Exploratory data analysis

Prior to the spatial autocorrelation analysis, an exploratory analysis was conducted. Total cases, projected population and annual incidence were computed at the national scale; descriptive statistics of provincial incidence were obtained by year, including the number of provinces with

no notified cases and the skewness coefficient; and the ten provinces with the highest cumulative incidence over the period were identified, computed as cumulative cases divided by the sum of annual projected populations. Annual choropleth maps were produced using fixed class intervals common to all years, in order to allow direct comparison between periods.

2.8 Spatial weights matrix

The application of spatial autocorrelation statistics requires a neighbourhood structure formalising the notion of proximity between geographical units. A 196×196 spatial weights matrix W was built using the Queen contiguity criterion, under which two provinces are considered neighbours when they share at least one boundary point (Anselin, 1988; Bivand et al., 2013).

The resulting structure connected all provinces without generating isolated units. The number of neighbours per province ranged from 1 to 11, with a mean of 5.36 and a median of 5. The same structure was used for all years of the period.

For Moran’s Index, both global and local, the matrix was row-standardised:

$$w_{ij} = \frac{c_{ij}}{\sum_{j=1}^n c_{ij}}, \quad \sum_{j=1}^n w_{ij} = 1, \quad (2)$$

where $c_{ij} = 1$ if provinces i and j are contiguous and $c_{ij} = 0$ otherwise, with $c_{ii} = 0$ by definition. Row standardisation allows the spatial lag to be interpreted as the average incidence observed in neighbouring provinces.

For the Getis–Ord G_i^* statistic, a second matrix was built, binary and including self-neighbourhood ($w_{ii} = 1$), in accordance with the original definition of the statistic (Getis and Ord, 1992; Ord and Getis, 1995). Additionally, for sensitivity analysis, a third matrix based on the five nearest neighbours (KNN, $k = 5$) was built, which guarantees a constant number of neighbours and allows assessment of the extent to which results depend on the contiguity criterion in a territory with provinces of highly heterogeneous area.

2.9 Variable transformation

The distribution of provincial incidence was markedly skewed in every year of the period, with skewness coefficients between 2.43 and 7.33, medians equal or close to zero, and maxima exceeding the median by two orders of magnitude. Between 46 % and 68 % of provinces notified no cases depending on the year.

Under these conditions, permutation tests applied to local statistics lose power, since the reference distribution is dominated by a small number of extreme values and most units present values tied at zero. For this reason, and following common practice in the spatial analysis of strongly skewed epidemiological rates, the statistics were computed on incidence transformed as $\log(\text{incidence} + 1)$. The unit constant allows provinces with no notified cases to be retained in the analysis. Results obtained on the original scale are reported as sensitivity analysis, so that the choice of scale can be transparently evaluated by the reader.

2.10 Global spatial autocorrelation: Moran's Index

To assess spatial dependence at the national scale, the Global Moran's Index was computed for each year (Anselin, 1988; Moran, 1950):

$$I = \frac{n}{\sum_{i=1}^n \sum_{j=1}^n w_{ij}} \cdot \frac{\sum_{i=1}^n \sum_{j=1}^n w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2}, \quad (3)$$

where $n = 196$, x_i is the transformed incidence of province i , $\bar{x}$ its mean and w_{ij} the corresponding element of the row-standardised matrix.

Positive and significant values indicate positive spatial autocorrelation; negative values indicate spatial dispersion; and values close to the expectation under the null hypothesis, $E[I] = -1/(n - 1) = -0.005128$, correspond to a spatially random distribution. Significance was assessed by Monte Carlo permutation with 9,999 permutations. A Moran scatterplot was also produced.

2.11 Cluster detection: the Getis-Ord G_i^* statistic

The local Getis-Ord G_i^* statistic was applied (Getis and Ord, 1992; Ord and Getis, 1995), which distinguishes clusters of high values from clusters of low values, a distinction the local Moran statistic does not provide directly:

$$G_i^* = \frac{\sum_{j=1}^n w_{ij} x_j - \bar{x} \sum_{j=1}^n w_{ij}}{S \sqrt{\frac{n \sum_{j=1}^n w_{ij}^2 - \left(\sum_{j=1}^n w_{ij}\right)^2}{n - 1}}}, \quad S = \sqrt{\frac{\sum_{j=1}^n x_j^2}{n} - \bar{x}^2}. \quad (4)$$

Unlike the Moran statistic, the weights matrix used includes the spatial unit itself ($w_{ii} = 1$), which is why the statistic is denoted with an asterisk. This specification is essential: if

the focal unit is excluded from the computation, the statistic reflects only the intensity of the neighbourhood and may classify provinces with no notified cases, surrounded by heavily affected provinces, as hotspots.

The result is interpreted as a Z score: positive values identify high-incidence clusters and negative values low-incidence clusters. Significance was assessed by conditional permutation (9,999) and the resulting p values were adjusted using the Benjamini–Hochberg false discovery rate.

The G_i^* statistic evaluates the concentration of the variable in the neighbourhood of each unit relative to the overall mean, and therefore identifies relative concentration rather than absolute disease burden. In years of low national incidence, a province may reach significance with a small number of notified cases. The results tables report absolute case counts alongside the value of the statistic so that this distinction is explicit for the reader.

2.12 Local spatial autocorrelation: LISA indicators

Local Indicators of Spatial Association proposed by Anselin (1995) were applied as a complementary analysis:

$$I_i = \frac{(x_i - \bar{x})}{m_2} \sum_{j=1}^n w_{ij}(x_j - \bar{x}), \quad m_2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}. \quad (5)$$

Based on the sign of the standardised value of the province and the sign of its spatial lag, significant results were classified into four categories: High–High (HH), Low–Low (LL), High–Low (HL) and Low–High (LH), the last two corresponding to spatial outliers.

Inference was based on conditional permutation with 9,999 permutations, and p values were adjusted using the Benjamini–Hochberg false discovery rate method with a significance level of $\alpha = 0.05$. For comparative purposes, and given that much of the literature on the spatial distribution of dengue uses the uncorrected criterion, the tables also report the unadjusted p value.

For both local statistics we used the pseudo p value based on the rank of the observation within the permutation distribution, a criterion equivalent to that implemented in GeoDa and adopted by most of the spatial epidemiology literature. It should be noted that this procedure imposes a lower bound on the attainable p value, equal to $1/(\text{permutations} + 1)$, which conditions the power of the multiple-comparison adjustment: when no observation attains a p value below

α/n , the Benjamini–Hochberg correction cannot reject any null hypothesis. For this reason, local results are reported under both inferential criteria.

Both local statistics were applied to every year of the 2018–2024 period. An indicator of temporal persistence was also constructed, defined as the number of years in which each province was classified as a significant cluster, in order to distinguish stable transmission foci from episodic ones.

2.13 Sensitivity analysis

To assess the robustness of the results, the Global Moran’s Index was recomputed under four specifications: (i) log-transformed incidence with Queen contiguity, used as the main analysis; (ii) incidence on the original scale with the same matrix; (iii) transformed incidence with a five-nearest-neighbours matrix; and (iv) rates smoothed by empirical Bayes estimation following the Assunção–Reis procedure, which corrects the excess sampling variability of provinces with small populations. The persistence analysis was also repeated restricted to the 2020–2024 subperiod, in which national incidence exceeded 130 cases per 100,000 inhabitants in every year.

2.14 Processing and software

All data processing, statistical analyses and map production were carried out in the R statistical programming environment (R Core Team, 2025). The packages `sf` were used for handling vector data, `spdep` for constructing the spatial weights matrices and computing the global and local autocorrelation statistics, `dplyr` and `tidyr` for data cleaning and integration, and `ggplot2` and `tmap` for generating figures and thematic maps. The cartography was processed in the WGS 84 geographic reference system (EPSG:4326), and the centroids used in the nearest-neighbour matrix were computed in UTM zone 18 South (EPSG:32718).

2.15 Ethical considerations

The study was based entirely on publicly available secondary databases, aggregated at the provincial level and containing no information permitting the identification of individuals. Owing to its ecological nature and the absence of direct human participation, it did not require approval by an institutional ethics committee or informed consent.

3 Results

3.1 Temporal behaviour of dengue, 2018–2024

Substantial temporal variation in provincial dengue incidence was recorded in Peru during 2018–2024. The spatial database analysed comprised 1372 observations corresponding to the 196 provinces of the country over seven consecutive years.

The annual evolution of notified cases showed a progressive increase, particularly from 2022 onwards. The highest case count was recorded in 2024, with 271,531 notifications and a national incidence of 797.72 cases per 100,000 inhabitants, whereas in 2018 there were 4,698 cases with an incidence of 14.88 (Table S1).

Two clearly differentiated stages are apparent. Between 2018 and 2021 national incidence remained below 150 cases per 100,000 inhabitants. From 2022 a sustained increase was observed, peaking in 2023 and 2024.

Table S2 presents descriptive statistics of provincial incidence by year. The distribution was markedly skewed in every year, with skewness coefficients between 2.43 and 7.33 and medians equal to zero up to 2023. The number of provinces with no notified cases fell progressively from 134 (68.4 %) in 2018 to 91 (46.4 %) in 2024, evidencing a sustained expansion of the geographical area affected by the disease.

Regarding the distribution of disease burden by department, Piura, Lima, La Libertad, Ica and Lambayeque concentrated the largest number of cumulative cases during the period. Examining cumulative incidence at the provincial scale (Table S3), the provinces with the highest rates corresponded mainly to the northern coast and the Amazon. Descriptive statistics confirmed marked spatial heterogeneity: while many provinces recorded very low or null incidence, a small group reached extremely high values.

3.2 Spatial distribution of dengue incidence

To examine the spatial evolution of dengue, choropleth maps were produced for each year of the period using fixed, common class intervals that allow direct comparison between years (Figure 1).

Marked spatial heterogeneity in disease intensity is apparent. The highest concentrations were located on the northern coast and in various provinces of the Peruvian Amazon, whereas much of the southern highlands showed low or null incidence. The provinces with the greatest burden form geographically continuous areas, particularly in Piura, La Libertad, Lambayeque, Tumbes,

Loreto, San Martín and Ucayali. The presence of contiguous high-incidence areas constitutes an initial indication of spatial dependence, formally assessed in the following sections.

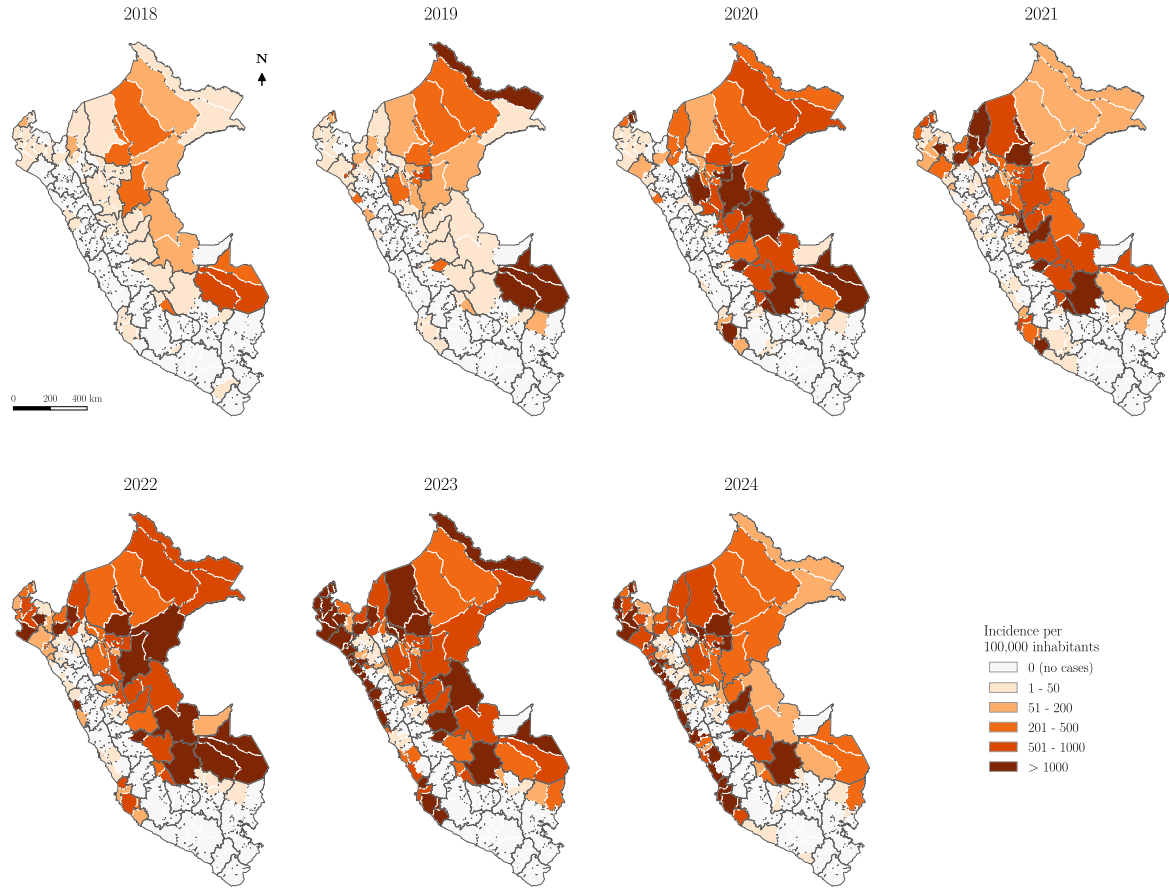


Figure 1: Spatial distribution of dengue incidence by province in Peru, 2018–2024. Class intervals are fixed and common to all years, allowing direct comparison of the evolution of both the intensity and the territorial extent of transmission. The lowest category corresponds to provinces with no notified cases.

3.3 Global spatial autocorrelation of dengue incidence

To quantitatively assess the spatial dependence of dengue incidence across the provinces of Peru, the Global Moran’s Index was computed for each year of the period. The values obtained were positive and statistically significant throughout (Table 2), indicating positive spatial autocorrelation in every year. Consequently, provinces with high incidence tended to be located close to other high-incidence provinces, while low-incidence provinces also showed a clustering pattern.

Table 2: Global Moran’s Index of dengue incidence in Peru, 2018–2024, and sensitivity analysis.

Year	I (log)	p	I (original scale)	I (KNN $k = 5$)	I (EB rates)	Skewness
2018	0.561	<0.001	0.308	0.490	0.307	6.29
2019	0.525	<0.001	0.258	0.505	0.258	7.33
2020	0.624	<0.001	0.439	0.553	0.439	2.88
2021	0.602	<0.001	0.299	0.562	0.299	2.80
2022	0.644	<0.001	0.281	0.610	0.281	3.19
2023	0.604	<0.001	0.612	0.611	0.613	2.43
2024	0.561	<0.001	0.285	0.579	0.285	3.54

Row-standardised Queen contiguity weights matrix; inference based on 9,999 Monte Carlo permutations. $E[I] = -0.005128$. The main column corresponds to incidence transformed as $\log(\text{incidence} + 1)$. EB: rates smoothed by empirical Bayes estimation (Assunção-Reis).

A central finding of the analysis is that the intensity of spatial dependence was remarkably stable over the period. On the logarithmic scale, the index ranged narrowly between 0.525 in 2019 and 0.644 in 2022, with no monotonic trend or discontinuity. The spatial structure of dengue in the Peruvian territory therefore remained essentially invariant over the seven years analysed, even across the change in epidemiological regime represented by the 2023 and 2024 epidemics.

This behaviour contrasts with that observed on the original incidence scale, where the index displays marked oscillation, with an isolated peak in 2023 ($I = 0.612$) and values close to 0.28 in adjacent years. The divergence between the two specifications is explained by the skewness of the distribution: on the original scale the statistic is strongly influenced by a small number of provinces with extreme incidence, whose magnitude varies from year to year. The agreement between the logarithmic specification and that based on the five-nearest-neighbour matrix, together with the near identity between the original scale and empirical Bayes smoothed rates, confirms that this divergence reflects the measurement scale and not the neighbourhood criterion or the small-numbers problem.

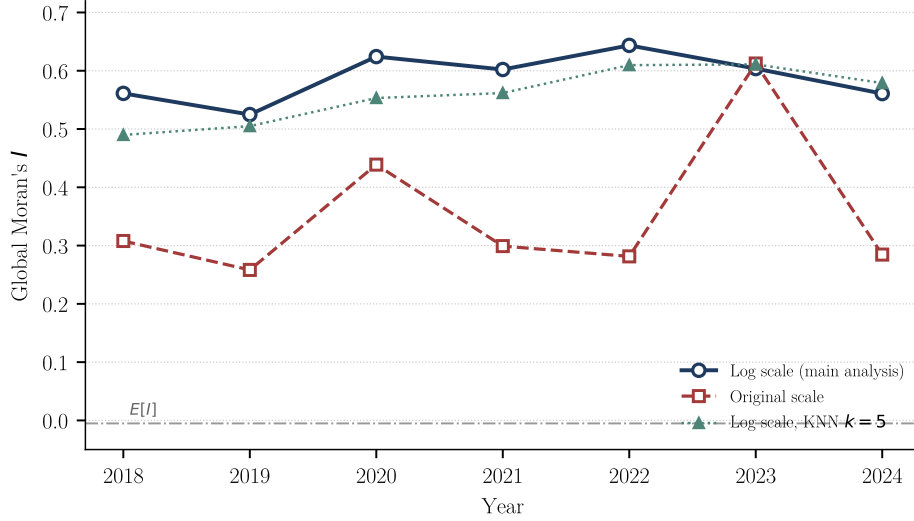


Figure 2: Temporal evolution of the Global Moran’s Index of dengue incidence in Peru, 2018–2024, under three specifications. On the logarithmic scale the index remains stable throughout the period, whereas on the original scale it displays an isolated peak in 2023 attributable to distributional skewness. The reference line indicates the expected value under the hypothesis of spatial randomness, $E[I] = -1/(n - 1)$.

The territorial expansion of transmission, by contrast, is evident when examining the geographical coverage of the disease: the number of provinces with no notified cases fell from 134 in 2018 to 91 in 2024, and the median provincial incidence departed from zero for the first time in 2024. It is this indicator, rather than the magnitude of Moran’s Index, that documents the progressive widening of the affected area.

The Moran scatterplot for 2024 (Figure S2) confirms the presence of positive spatial autocorrelation and justifies the application of local indicators to identify the specific location of clusters.

3.4 Cluster identification using the Getis–Ord G_i^* statistic

In order to locate areas of significant concentration and to distinguish explicitly between hot and cold areas, the local Getis–Ord G_i^* statistic was applied to every year of the period. Table S4 summarises the annual results under both inferential criteria and Table 3 details the hotspots identified in 2024.

The number of hotspots ranged between 20 and 35 provinces depending on the year, and that of coldspots between 31 and 61, without the multiple-comparison correction substantially altering these magnitudes. In 2024, 23 provinces were identified as hotspots and 31 as coldspots.

The 2024 hotspots were distributed across Piura (5 provinces), San Martín (6), Loreto (4),

Cajamarca (2), La Libertad (2), and Lambayeque, Lima and Tumbes (1 each). The highest values of the statistic corresponded to Sullana, Lambayeque, Piura and Paita, on the northern coastal axis, and to Alto Amazonas and Lamas, on the Amazonian axis. The province with the fewest cases among the hotspots recorded 87 notifications, confirming that the classification identifies territories with effective transmission.

Table 3: Provinces identified as significant dengue incidence hotspots by the Getis–Ord G_i^* statistic (2024).

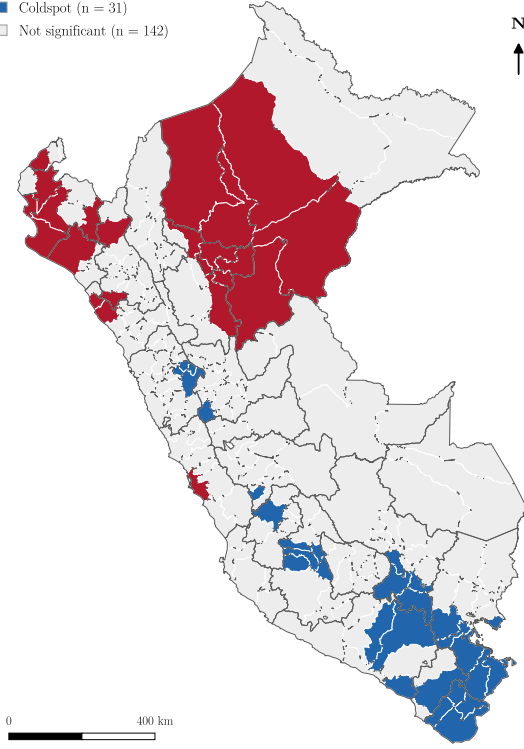
Province	Department	Cases	Incidence ^a	G_i^*	p_{FDR}
Sullana	Piura	2,744	775.6	3.454	0.004
Lambayeque	Lambayeque	2,529	692.3	3.408	0.007
Piura	Piura	14,580	1,520.2	3.383	0.009
Paita	Piura	5,087	3,178.9	3.351	0.006
Alto Amazonas	Loreto	4,497	2,811.1	3.318	0.014
Lamas	San Martin	612	677.6	3.091	0.014
Moyobamba	San Martin	2,395	1,740.3	3.047	0.022
Ascope	La Libertad	5,946	4,722.9	2.941	0.039
Contumaza	Cajamarca	366	1,292.4	2.898	0.032
Sechura	Piura	2,527	2,564.5	2.888	0.039
Datem Del Marañon	Loreto	517	840.2	2.874	0.026
Pacasmayo	La Libertad	1,798	1,520.8	2.839	0.028
Requena	Loreto	238	374.9	2.797	0.034
San Martin	San Martin	4,388	1,878.6	2.780	0.039
Lima	Lima	74,040	719.4	2.699	0.038
Contralmirante Villar	Tumbes	87	328.0	2.674	0.024
Jaen	Cajamarca	1,773	859.2	2.633	0.044
Bellavista	San Martin	281	439.1	2.620	0.039
El Dorado	San Martin	239	525.1	2.585	0.042
Picota	San Martin	100	219.9	2.582	0.037
Ucayali	Loreto	142	225.7	2.574	0.040
Loreto	Loreto	202	272.9	2.522	0.043
Huancabamba	Piura	89	76.0	2.327	0.043

^a Cases per 100,000 inhabitants in 2024. Binary Queen contiguity weights including self-neighbourhood ($w_{ii} = 1$); 9,999 conditional permutations; Benjamini–Hochberg false discovery rate correction. The 31 coldspots are listed in Appendix A.1.

Coldspots were concentrated entirely in the southern and central highlands: Puno (7 provinces), Ayacucho (5), Áncash (4), Cusco (4), Tacna (4), Arequipa (3), and Huancavelica, Huánuco, Junín and Moquegua (1 each). Their identification allows positive delimitation of the area of the national territory where sustained transmission is unlikely under current ecological conditions.

A. Getis-Ord G_i^* clusters, 2024

Hotspot (n = 23)
Coldspot (n = 31)
Not significant (n = 142)



B. Hotspot persistence, 2018–2024

Years
0 1 2 3 4 5 6 7

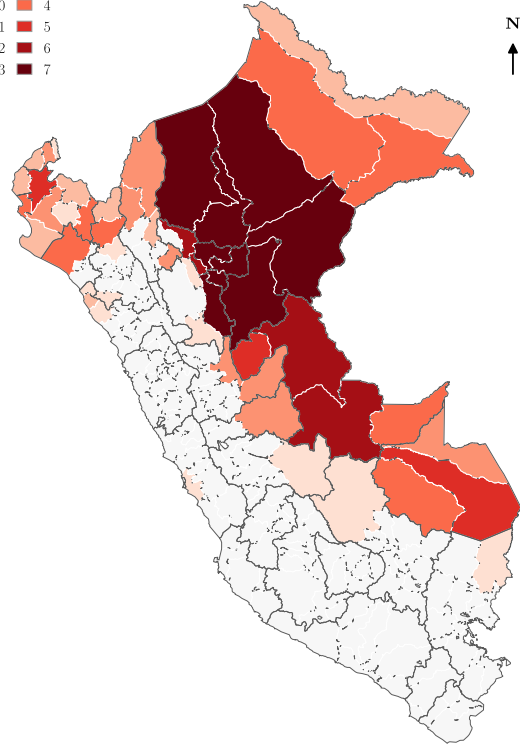


Figure 3: Spatial clusters of dengue incidence in Peru identified by the Getis-Ord G_i^* statistic. (A) Hot and cold clusters in 2024. (B) Temporal persistence, 2018–2024: the value for each province is the number of years in which it was classified as a statistically significant hotspot. Binary Queen contiguity weights with self-neighbourhood; inference by conditional permutation (9,999) with Benjamini–Hochberg correction.

3.5 Complementary analysis using Local Indicators of Spatial Association

Local Indicators of Spatial Association were applied as a complementary analysis. The results, presented in Table S5, differ markedly from those obtained with G_i^* and merit specific consideration.

Under the uncorrected criterion, the number of High–High clusters increased from 28 in 2018 to 41 in 2023, standing at 35 in 2024, and that of Low–Low clusters grew from 5 to 39 over the same period. Under the FDR-corrected criterion, by contrast, the number of High–High clusters ranged between 10 and 25 up to 2023 and fell to zero in 2024.

This absence of significance in the final year does not reflect an absence of clustering but a limitation of the inferential procedure. The minimum p value attained in 2024 was 7×10^{-4} , above the 2.55×10^{-4} threshold that the Benjamini–Hochberg correction requires for the first rejection

with 196 simultaneous tests; no province could therefore retain significance under that criterion. Table S6 lists the High–High provinces identified in 2024 under the uncorrected criterion, with their p values under both versions, and Figure S3 maps their distribution.

The divergence between the two statistics is informative. The G_i^* statistic compares the intensity of each province’s neighbourhood with the national mean, whereas the local Moran statistic evaluates the covariance between each province and its neighbours. During 2024 transmission reached its greatest territorial extent of the period, with only 91 provinces reporting no cases compared with 134 in 2018, and this homogenisation reduced the local contrast required by the Moran statistic without affecting the concentration that G_i^* detects. For this reason the cluster analysis in the present study relies principally on the G_i^* statistic.

3.6 Temporal persistence of clusters

Applying the G_i^* statistic to the whole period allowed the temporal stability of transmission foci to be assessed (Table 4; Figure 3B). Nine provinces were classified as hotspots in all seven consecutive years: Alto Amazonas, Datem del Marañón, Loreto, Requena and Ucayali, in the department of Loreto; and Bellavista, Lamas, Picota and San Martín, in San Martín. A further four reached six years — El Dorado and Moyobamba, in San Martín, and Atalaya and Coronel Portillo, in Ucayali — and three reached five: Tambopata (Madre de Dios), Padre Abad (Ucayali) and Sullana (Piura).

Table 4: Provinces with hotspot persistence in five or more years of the 2018–2024 period.

Province	Department	Years (2018–2024)	Years (2020–2024)	2024 incidence ^a
Alto Amazonas	Loreto	7	5	2,811.1
Bellavista	San Martin	7	5	439.1
Datem Del Marañon	Loreto	7	5	840.2
Lamas	San Martin	7	5	677.6
Loreto	Loreto	7	5	272.9
Picota	San Martin	7	5	219.9
Requena	Loreto	7	5	374.9
San Martin	San Martin	7	5	1,878.6
Ucayali	Loreto	7	5	225.7
El Dorado	San Martin	6	5	525.1
Moyobamba	San Martin	6	5	1,740.3
Atalaya	Ucayali	6	4	69.2
Coronel Portillo	Ucayali	6	4	86.0
Padre Abad	Ucayali	5	4	323.7
Sullana	Piura	5	4	775.6
Tambopata	Madre De Dios	5	3	271.3

^a Cases per 100,000 inhabitants. The 2020–2024 column restricts the analysis to years of substantial national incidence and is presented as a robustness check.

Since national incidence was considerably lower in 2018 and 2019 and some of these provinces recorded few absolute cases, the analysis was repeated restricted to 2020–2024, a period in which national incidence exceeded 130 cases per 100,000 inhabitants in every year. Eleven provinces were hotspots in all five years: five in Loreto and six in San Martín. The identified structure therefore does not depend on the low-incidence years.

This result reveals a configuration of two clearly differentiated components. On the one hand, an endemic Amazonian core, comprising provinces of Loreto, San Martín, Ucayali and Madre de Dios, which maintained hotspot status independently of the national epidemic cycle. On the other, a coastal component of episodic character: with the sole exception of Sullana, no coastal province reached five or more years of persistence, despite concentrating the highest values of the statistic during 2023 and 2024.

It should be noted that in 2023 and 2024 only four and six, respectively, of the thirteen provinces of the Amazonian core exceeded the national incidence, a consequence of the high level the latter reached during the coastal epidemics. The Amazonian core therefore constitutes a persistent focus of spatial concentration rather than the territory with the highest absolute incidence in every year.

4 Discussion

4.1 Interpretation of the main findings

The results show that dengue incidence in Peru increased steadily during 2018–2024, reaching its greatest magnitude in 2023 and 2024. In parallel, the spatial analyses demonstrated that the geographical distribution of the disease did not follow a random pattern but presented a significantly clustered and, as the central finding of this work, remarkably stable spatial structure over time.

The persistence of positive and statistically significant values of the Global Moran’s Index throughout the period demonstrates positive spatial autocorrelation in dengue incidence. In epidemiological terms, this indicates that the situation of a province is conditioned by the conditions present in neighbouring provinces, as described in reference work on spatial statistics (Anselin, 1995; Bivand et al., 2013; Moran, 1950).

One aspect deserving specific attention is that the magnitude of this spatial dependence remained within a narrow range across the seven years analysed, despite the substantial change in epidemiological regime represented by the 2023 and 2024 epidemics. This result qualifies an interpretation frequently found in the literature, according to which interannual variations in Moran’s Index reflect changes in the territorial organisation of transmission. In the present study, the oscillation observed on the original incidence scale — with an isolated peak in 2023 and considerably lower values in adjacent years — disappeared when a logarithmic transformation was applied, and was therefore attributable to distributional skewness rather than to a geographical reorganisation of the disease. Since Moran’s Index is built on deviations from the mean, in strongly skewed distributions a small number of provinces with extreme incidence can dominate the statistic, so that its value reflects the magnitude of each year’s outliers rather than the underlying spatial structure.

This observation has a methodological implication beyond the Peruvian case: the substantive interpretation of temporal variations in Moran’s Index requires prior verification of its robustness to the measurement scale. The agreement between the logarithmic specification and that based on the five-nearest-neighbour matrix, together with the near identity between the original scale and empirical Bayes smoothed rates, rules out both the neighbourhood criterion and the small-numbers problem as alternative explanations for this divergence.

The territorial expansion of the disease, by contrast, is unequivocally manifest in the geo-

graphical coverage of notification: the number of provinces with no recorded cases fell from 134 to 91 between 2018 and 2024, and the median provincial incidence departed from zero for the first time in 2024. The disease did not become more concentrated; it reached a progressively larger number of territories while maintaining its pattern of spatial organisation.

4.2 An endemic Amazonian core and an episodic coastal expansion

Applying the local statistics to the whole period allowed the identification of a two-component spatial structure with clearly differentiated dynamics.

The first component corresponds to an endemic Amazonian core. Nine provinces of the departments of Loreto and San Martín were classified as hotspots in all seven consecutive years of the period, and eleven provinces of those same departments were hotspots in all five years of the 2020–2024 subperiod, when national incidence was substantial in every year. This core maintained its status independently of the national epidemic cycle. Its stability is compatible with the ecological conditions of the Peruvian Amazon, where temperature and humidity allow year-round activity of *Aedes aegypti*, in contrast with the marked seasonality of the coast.

The second component corresponds to an episodic coastal expansion. The provinces of the northern and central coast reached their greatest extent as significant clusters during 2023, when all provinces of Tumbes and five provinces of Piura, together with provinces of Lambayeque, Cajamarca and La Libertad, formed a continuous block of intense transmission. With the sole exception of Sullana, no coastal province reached five or more years of persistence during the period. This component therefore reflects a recent expansion process rather than a historically established focus.

The distinction between the two components is directly relevant to the interpretation of the 2023 epidemic, the largest documented in the country’s recent history. The results suggest that this epidemic did not consist of a uniform intensification of national transmission but of the transient incorporation of a large coastal territory into a clustering pattern that the Amazon had been sustaining permanently. The climatic conditions associated with the coastal El Niño phenomenon, widely documented as determinants of breeding-site proliferation on the northern Peruvian coast, offer a plausible explanation for the episodic nature of this component (Casapía-Morales et al., 2023; Rufasto Goche et al., 2025).

The coldspots identified by the G_i^* statistic reinforce this reading. They were located entirely in the southern and central highlands — Puno, Ayacucho, Cusco, Tacna, Arequipa, Huancavelica,

Huánuco, Junín and Moquegua — territories whose altitude imposes an ecological limit on vector distribution. The identification of these clusters, absent from previous analyses conducted on the original incidence scale, allows positive delimitation of the area of the national territory where sustained transmission is unlikely.

4.3 Comparison with national and international research

The spatial patterns identified are consistent with research conducted both in Peru and in other dengue-endemic countries. In the Peruvian context, Rufasto Goche et al. (2025) analysed the epidemiological dynamics of dengue between 2000 and 2022 and concluded that departments such as Loreto, Ucayali, Piura and San Martín persistently concentrated the highest disease burden. The results of the present study agree with those findings at the provincial scale and add further precision: they allow a distinction between which of those territories constitute permanent foci — the Amazonian provinces of Loreto, San Martín and Ucayali — and which are incorporated episodically during epidemic cycles, as occurs with the coast of Piura, Tumbes and Lambayeque.

More recently, Burgos et al. (2025) examined the spatiotemporal dynamics of dengue in Amazonas, north-eastern Peru, between 2000 and 2023, reporting sustained transmission associated with climatic variability. The present study extends that observation to the national scale and shows that the Amazonian pattern is not confined to a single department but constitutes a persistent core spanning Loreto, San Martín, Ucayali and Madre de Dios.

Similarly, the studies conducted in Iquitos by Morrison et al. (2010) and Liebman et al. (2012) demonstrated that dengue transmission exhibits marked spatial heterogeneity even at local scales, identifying persistent foci associated with vector distribution and population mobility. The seven-year persistence documented here for the provinces of Loreto is consistent with those findings and suggests that the pattern described at the intra-urban scale in Iquitos is reproduced at the provincial scale across the Peruvian Amazon.

Internationally, the results are also consistent with research conducted in Brazil, where Mondini and Chiaravalloti Neto (2008) identified positive spatial autocorrelation of dengue incidence using the Global Moran’s Index, showing that the disease concentrates in geographical areas with similar socioeconomic and environmental characteristics. Likewise, Stewart-Ibarra et al. (2014) reported high-incidence spatial clusters in Ecuador, associated mainly with high temperatures, precipitation and urbanisation processes.

Similar results have recently been reported in Mexico, Nepal, Bangladesh and Singapore,

where the joint application of the Global Moran’s Index, LISA indicators and the Getis–Ord G_i^* statistic allowed the identification of persistent spatial clustering patterns and priority areas for surveillance (Gurram et al., 2025; Mahato et al., 2025; Mendoza-Cano et al., 2025; Sharif et al., 2022). The scoping review by Prakash Nayak et al. (2025) concludes that these three statistics are currently the most widely used tools for studying the spatial distribution of dengue worldwide, so that the methodology developed here is aligned with the most recent recommendations in the literature.

It should be noted, however, that most of these studies apply local statistics to a single year or to the cumulative incidence of the period, and on the original scale of the variable. The present study suggests that both decisions may substantively condition the conclusions: analysing a single year prevents permanent foci from being distinguished from episodic ones, and working on the original scale may attribute to changes in spatial structure variations that respond to distributional skewness.

4.4 Public health implications

The findings have relevant implications for epidemiological surveillance and health management in Peru. The identification of two spatial components with distinct dynamics suggests that a uniform territorial strategy would be inadequate, and that interventions should be differentiated according to the type of focus.

In the endemic Amazonian core, where hotspot status was maintained without interruption over the seven years analysed, priority lies with sustained vector control and permanent entomological surveillance rather than with emergency responses triggered by outbreaks. The provinces of Loreto, San Martín, Ucayali and Madre de Dios identified in the persistence analysis are natural candidates for continuous intervention programmes.

In the coastal component, episodic in character and associated with favourable climatic conditions, priority lies with developing rapid response capacity and early warning systems linked to the monitoring of climatic variables, particularly events associated with the El Niño phenomenon. The rapid incorporation of the northern coast into the clustering pattern during 2023 and its subsequent partial retraction in 2024 illustrate the speed with which these territories can move between contrasting epidemiological conditions.

The delimitation of coldspots in the southern and central highlands additionally provides a criterion for efficient resource allocation, by positively identifying territories where sustained

transmission is unlikely under current ecological conditions. This delimitation should nevertheless be interpreted as a snapshot of the period analysed and not as a permanent guarantee, given that the altitudinal ascent of the vector documented in other Andean regions could modify it.

Finally, periodic updating of these analyses would allow early detection of changes in the geographical distribution of the disease and identification of new areas of expansion. This information is especially relevant in a context of climate change, accelerated urbanisation and increasing population mobility (Casapía-Morales et al., 2023; Prakash Nayak et al., 2025; World Health Organization, 2024).

4.5 Strengths and limitations

The main strength of this research lies in the integration of official epidemiological information, population projections and provincial cartography within a Geographic Information System, which allowed the construction of a balanced panel covering all 196 provinces of the country over seven consecutive years. Applying the local statistics to every year of the period, rather than only to the year of highest incidence, made it possible to distinguish permanent from episodic foci, a distinction that constitutes the main contribution of this work and is not attainable through cross-sectional analyses.

A second strength is the systematic incorporation of sensitivity analyses. Evaluating Moran’s Index under four specifications made it possible to identify the temporal variations observed on the original scale as a distributional artefact, and to rule out the neighbourhood criterion and the small-numbers problem as alternative explanations. The modular workflow adopted, which keeps the original data unaltered and documents each stage in executable scripts with a fixed random seed, ensures the reproducibility of the analysis.

Several limitations should nevertheless be considered. First, the analysis used information aggregated at the provincial level, so smaller-scale spatial variations may not have been captured; the patterns identified represent general trends and do not necessarily reflect the heterogeneity existing within each province. As an ecological study, the conclusions are not extrapolable to the individual level.

Second, the logarithmic transformation adopted as the main specification modifies the interpretation of the statistics, which come to reflect clustering of the order of magnitude of incidence rather than of its absolute value. This decision is justified by the skewness of the distribution and its consequences for the power of permutation tests, and its effects are transparently documented

in the sensitivity analysis, but it conditions direct comparability with studies operating on the original scale.

Third, a limitation of the inferential procedure must be considered. The pseudo p value obtained by permutation has a lower bound determined by the number of simulations, which in the case of the local Moran statistic proved insufficient for the multiple-comparison correction to reject any null hypothesis in 2024. This behaviour does not indicate an absence of clustering but a limitation of power, and has been explicitly flagged in the presentation of results. A larger number of permutations would reduce this bound, at the cost of considerably greater computation time.

Fourth, the G_i^* statistic identifies relative concentration rather than absolute disease burden. During the years of lowest national incidence, six observations corresponding to provinces with no notified cases reached significance as hotspots, being surrounded by territories with comparatively high incidence within a national context of low transmission. None of these observations corresponds to the provinces forming the persistent core identified, and all are concentrated in the 2018–2022 subperiod.

Fifth, the data come from a passive surveillance system subject to under-notification and to differences in diagnostic capacity between regions. It is plausible that limited access to health services in dispersed Amazonian areas reduces case registration, in which case the true magnitude of transmission in the endemic core identified would be underestimated.

Finally, the study focused on the descriptive analysis of spatial distribution and did not incorporate explanatory statistical models allowing quantification of the effect of climatic, environmental, socioeconomic or demographic variables on incidence. The results should therefore be interpreted as evidence of spatial patterns and not as causal relationships.

4.6 Future perspectives

The results obtained provide a basis for more advanced studies of the spatio-temporal dynamics of dengue in Peru. Future research could incorporate spatial regression models — spatial autoregressive models, spatial error models or geographically weighted regression — in order to quantify the influence of environmental, climatic, demographic and socioeconomic variables on the distribution of the disease. The distinction between the endemic core and the episodic component identified here offers a testable hypothesis: the determinants of transmission would be expected to differ between the two components, with a greater weight of interannual climatic

variables in the coastal component.

It would also be advisable to integrate high-resolution meteorological information, including temperature, precipitation, relative humidity and extreme climatic events, as well as satellite imagery and vegetation indices, in order to characterise more precisely the environmental conditions associated with the persistence of the identified foci.

The development of spatio-temporal models integrating both dimensions simultaneously would allow assessment of the spread of the disease between provinces, identification of transmission corridors and estimation of the risk of new outbreaks. The application of hierarchical Bayesian models, particularly through INLA approximations with conditional autoregressive spatial structures, constitutes a promising alternative for modelling case counts directly, avoiding the need to transform the response variable.

Finally, the integration of machine learning techniques with Geographic Information Systems and epidemiological, climatic and environmental data would allow the development of early warning systems capable of anticipating outbreaks with greater precision, contributing to optimising resource allocation and strengthening national dengue prevention and control strategies.

5 Conclusions

Dengue incidence in Peru increased steadily during 2018–2024, rising from 14.88 to 797.72 cases per 100,000 inhabitants between the first and last years analysed, with the highest values in 2023 and 2024. This increase was accompanied by an expansion of the geographical area affected: the number of provinces with no notified cases fell from 134 to 91, and the median provincial incidence departed from zero for the first time in 2024.

The spatial analyses demonstrated that the distribution of dengue was not random but presented statistically significant spatial dependence in every year of the period. Contrary to what analyses conducted on the original incidence scale would suggest, the intensity of this dependence remained stable, with no variation attributable to the epidemic cycle. The apparent oscillation of Moran’s Index proved to be an artefact of distributional skewness, a result that underlines the need to verify the robustness of this statistic to the measurement scale before attributing epidemiological meaning to its temporal variations.

Applying the local indicators to the whole period allowed the identification of two spatial components with differentiated dynamics. An endemic Amazonian core, comprising provinces

of Loreto, San Martín, Ucayali and Madre de Dios, maintained hotspot status almost without interruption over the seven years, with nine provinces significant throughout the period and eleven significant in all five years of the 2020–2024 subperiod. A coastal component, by contrast, reached its greatest extent during the 2023 epidemic and lacks historical persistence. Coldspots were located entirely in the southern and central highlands, delimiting the area where sustained transmission is unlikely under current ecological conditions.

This distinction between permanent and episodic foci is not identifiable through cross-sectional analyses and constitutes the main contribution of the present work. Its public health implications are direct: the endemic core requires sustained vector control and permanent entomological surveillance, whereas the coastal component calls for rapid response capacity and early warning systems linked to climate monitoring.

The methodology developed provides a replicable basis for future studies of the spatio-temporal dynamics of dengue in Peru, particularly those incorporating climatic and environmental variables through spatial regression models or hierarchical Bayesian approaches.

Declarations

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Author contributions

The author participated in the conception and design of the study, data collection and processing, analysis and interpretation of results, drafting of the manuscript and approval of its final version.

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Declaration of competing interest

The author declares no competing interests in relation to this research.

Ethics approval

The study was based entirely on publicly available secondary databases, aggregated at the provincial level and containing no information permitting the identification of individuals. Owing to its ecological nature and the absence of direct human participation, it did not require approval by an institutional ethics committee or informed consent.

Data availability

The databases used are publicly available at the National Open Data Portal of Peru and at the institutional portal of the National Institute of Statistics and Informatics. The processing and analysis scripts developed for this study, together with the derived spatial database and the numerical results, are deposited in the public Zenodo repository and available at <https://doi.org/10.5281/zenodo.21773936>.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used Claude (Anthropic) to assist with drafting and language editing of the English text, to review the statistical analysis code, and to verify bibliographic references. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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